Deriving Expressions and Formulae

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

A+ Maths Hub

Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 64 minutes · Total marks: 64

KEY RULES

Define letters	Let x = the unknown quantity
Translate	Turn the words into an expression, then into an equation
Perimeter / area	Write expressions for the sides, then use the formula
Solve and check	Solve the equation and check the answer fits the problem

WORKED EXAMPLE 1

[3 marks]

A rectangle has length $(x + 3)$ cm and width 5 cm. The perimeter is 36 cm. Find x.

Step 1	Write an equation for the perimeter	$2(x + 3 + 5) = 36$
Step 2	Expand and simplify	$2x + 16 = 36$
Step 3	Solve	$2x = 20, \quad x = 10$

Answer: $x = 10$

WORKED EXAMPLE 2

[3 marks]

A pen costs x pence. A ruler costs 20 pence more than a pen. 3 pens and 1 ruler cost 100 pence. Find the c

Step 1	Cost of a ruler	$x + 20$
Step 2	Write an equation	$3x + x + 20 = 100$
Step 3	Solve	$4x = 80, \quad x = 20$

Answer: 20 **pence**

WORKED EXAMPLE 3

[3 marks]

The sum of three consecutive integers is 78. Find the integers.

Step 1	Let the integers be $n,\ n + 1,\ n + 2$	$n + (n + 1) + (n + 2) = 78$
Step 2	Simplify and solve	$3n + 3 = 78, \quad n = 25$

Answer: 25, 26, 27

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1 Ali is x years old. Beth is 5 years older than Ali.

Cara is twice as old as Ali. The sum of their ages is 61.

Work out Ali's age.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

2 12 tiles and one bag of adhesive costing 30 dollars cost 90 dollars altogether.

Each tile costs x dollars. Work out x.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

3 Alisa, Jena and Mikael each pick cucumbers.

Alisa picks C cucumbers.
Jena picks 5 fewer cucumbers than Alisa.
Mikael picks twice as many cucumbers as Alisa.

The total number of cucumbers picked by Alisa, Jena and Mikael is T

Find a formula for T in terms of C
Give your formula in its simplest form.

.....

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 12

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4 The diagram below shows the trapezium *PQRS*

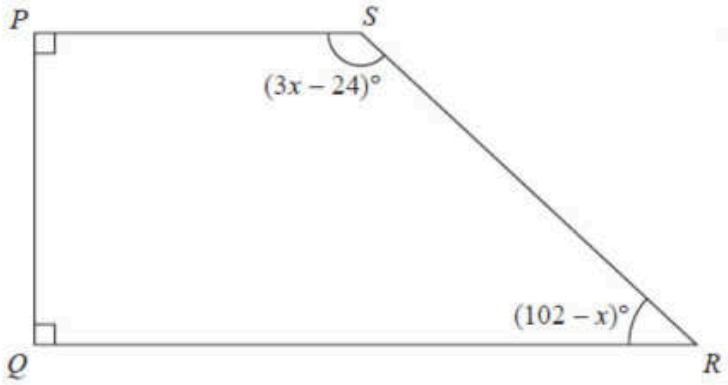


Diagram **NOT** accurately drawn

Angle *PQR* and angle *QPS* are right angles.

Find the value of *x*

x = ...

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2022 · Question 12

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5 Kye has some toy cars.

He has n red cars.
He has twice as many blue cars as red cars.
He has 7 more green cars than red cars.

(a) Write an expression, in terms of n, for the total number of red cars, blue cars and green cars that Kye has.

Write your answer in its simplest form.

.....

(2)

The total number of cars that Kye has is T

(b) Write an expression, in terms of T and n, for the number of cars that Kye has that are **not** red cars, blue cars or green cars.

.....

(1)

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2024 · Question 13

6 A small bag of grain contains 2 kg of grain.
A large sack of grain contains 7 kg of grain.

Makenna buys r small bags of grain and h large sacks of grain.

The total amount of grain Makenna buys is F kg.

Write down a formula for F in terms of r and h

.....

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2023 · Question 14

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7 There are 6 batteries in a small packet of batteries.
There are 9 batteries in a large packet of batteries.

Chow buys m small packets of batteries and g large packets of batteries.

The total number of batteries Chow buys is T.

Write down a formula, in terms of m and g, for T.

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2017 · Question 6

8 OAB is a sector of a circle with centre O and radius r cm.

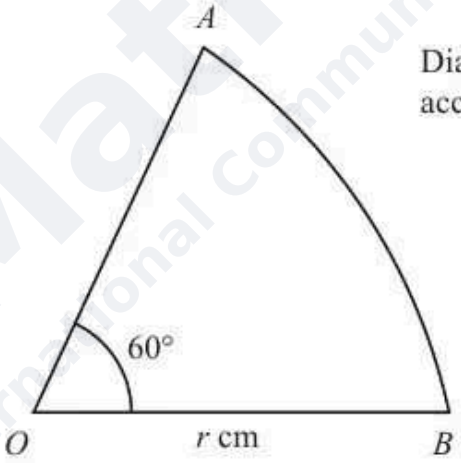


Diagram **NOT** accurately drawn

Angle $AOB = 60°$
The perimeter of the sector is P cm.

Find a formula for P in terms of r
Give your answer in the form $P = r(c\pi + k)$ where c and k are values to be found.

(Total for Question 13 is 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 13

9 A rectangle has length 3 cm more than its width w cm. Its area is 54 cm^2.

(a) Show that $w^2 + 3w - 54 = 0$

(b) Work out the width of the rectangle.

(Total for question = 4 marks)

Practice question written for this worksheet (not from a past paper)

10 A rectangle has length $(2x + 1)$ cm and width $(x + 4)$ cm.

Write expressions, in simplest form, for (a) its area (b) its perimeter.

(Total for question = 4 marks)

Practice question written for this worksheet (not from a past paper)

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11 450 students were asked how they travelled to school on Monday.
Each student walked or travelled by bus or travelled by car or travelled by bicycle.
Each student used just one method of travel.

One of these students is chosen at random.
The table shows information about the probability of each method of travel.

Method of travel	walk	bus	car	bicycle
Probability	0.20	x	$2x$	0.26

Work out how many of the 450 students travelled by car.

.....

(Total for question = 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 2

12 Sid has n counters.
Erin has 7 more counters than Sid has.
Kabir has 3 times as many counters as Sid has.

(a) Write down an expression, in terms of n, for the total number of counters that Sid, Erin and Kabir have.

.....

(2)

Kabir has 72 counters.

(b) How many counters does Erin have?

.....

(2)

(Total for question = 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2025 · Question 15

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CHALLENGE QUESTIONS

13

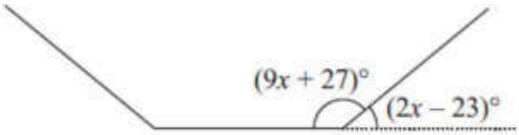


Diagram **NOT** accurately drawn

The diagram shows part of a regular n-sided polygon with

an interior angle of $(9x + 27)°$
an exterior angle of $(2x - 23)°$

Work out the value of n

n = ...

(Total for question = 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 12

14 Here are the first three terms of an arithmetic sequence.

$$(4x - 14) \quad , \quad (x + 2) \quad , \quad (7x - 9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

.....

(Total for question = 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 23

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15 A bag contains x marbles.
12 of the marbles are pink and the rest are yellow.

Greta takes at random 2 marbles from the bag.

The probability that Greta takes a pink marble and a yellow marble is $\frac{18}{47}$

Work out the value of x
Show clear algebraic working.

$x =$ .....

(Total for question = 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 23

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16

23

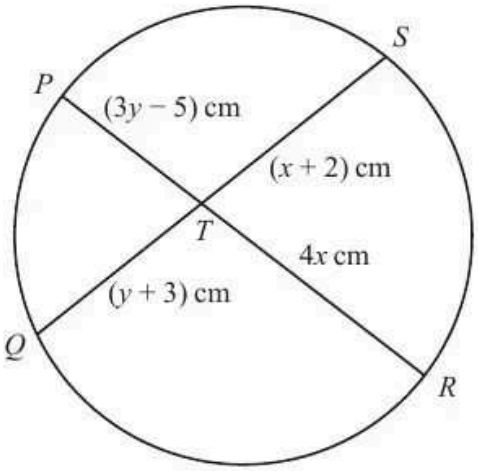


Diagram **NOT** accurately drawn

PTR and *QTS* are chords of a circle.

$PT = (3y - 5)\,\text{cm}$ $QT = (y + 3)\,\text{cm}$ $RT = 4x\,\text{cm}$ $ST = (x + 2)\,\text{cm}$

Find an expression for y in terms of x

$y =$ .....

Total for Question 23 is 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 23

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17 The first 3 terms of an arithmetic series are

$$(2x + 5) \quad (3y - 4) \quad (4x - 2)$$

where x and y are constants.

The sum of the first 9 terms of the series is 216

Find the value of x and the value of y
Show clear algebraic working.

$x =$ .....

$y =$ .....

(Total for question = 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 24

TOTAL FOR PAPER IS 64 MARKS

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